

1. Administrative & Study Identification

Full Study Title: A Phase 1b/2 Study of BMS-986158 Monotherapy and in Combination With Either Ruxolitinib or Fedratinib in Participants With DIPSS-Intermediate or High Risk Myelofibrosis
Short Title/Acronym: CA011-023 **Protocol Number:** CA011-023 **NCT Number:** NCT04817007 **Sponsor Name:** Bristol Myers Squibb (Company: Bristol Myers Squibb) **Investigational Product:** BMS-986158 (BET Inhibitor) alone and in combination with Ruxolitinib (Trade names: Jakafi, Opzelura) or Fedratinib (JAK inhibitors) **Phase of Development:** PHASE1, PHASE2 **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety, tolerability, maximum tolerated dose (MTD) and/or recommended Phase 2 dose (RP2D) of BMS-986158 alone and in combination with either Ruxolitinib or Fedratinib. **Secondary Objectives:** To evaluate preliminary efficacy based on Spleen Volume Reduction (SVR), symptom improvement (Total Symptom Score), pharmacokinetic (PK) parameters, and overall survival (OS).

2.2 Background & Rationale An unmet medical need remains in Myelofibrosis (MF) for new treatments that lead to strong and durable spleen volume reduction, symptom improvement, and extended survival. Bromodomain and extraterminal (BET) inhibitors alone and in combination with Janus kinase (JAK) inhibitors have shown clinical benefit in patients with MF. Concomitant inhibition of BET-mediated and JAK/STAT pathways is hypothesized to have additional therapeutic benefits over BET-targeted inhibition alone.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Trial Status:** ACTIVE_NOT_RECRUITING **Control Type:** Single-arm/Sequential Assignment **Blinding:** Open-label (None) **Randomization:** N/A (Sequential assignment based on part and cohort)

3.2 Planned Enrollment & Duration Target \$N\$: 216 participants
Number of Sites: Multicenter, global clinical trial sites
Estimated Study Start: 22-Mar-2021 **Estimated Primary Completion:** 31-May-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Diagnosis of primary myelofibrosis (PMF), post-essential thrombocythemia (ET) or post-polycythemia vera (PV) myelofibrosis. 2. Dynamic International Prognostic Scoring System (DIPSS) intermediate or high-risk blood cancer. 3. Treatment-related toxicities from prior therapy resolved to Grade 1 or pre-treatment baseline or determined to be irreversible prior to study treatment. 4. Age ≥ 18 years. 5. Must agree to follow specific methods of contraception, if applicable.

4.2 Exclusion Criteria 1. Women who are pregnant or breastfeeding at screening. 2. Any significant acute or uncontrolled chronic medical illness.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered based on sequential dose-escalation and expansion rules. Part 1 consists of BMS-986158 in combination with either Ruxolitinib (Part 1A) or Fedratinib (Part 1B). Part 2 consists of BMS-986158 in combination with either Ruxolitinib or Fedratinib and BMS-986158 alone. **Route of Administration:** Oral **Duration of Treatment:** Continued until disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for symptoms. **Prohibited:** Concurrent investigational therapies and specific strong CYP modulators depending on specific combination arm parameters.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Day -28 to 0	Informed Consent, Medical History, Baseline DIPSS Scoring
2	Day 0	Treatment Assignment, First Dose
3-7	Defined Cycle Weeks	Safety assessments, PK blood draws, Spleen Volume imaging (CT/MRI), Symptom Questionnaires
8	Follow-up	Final safety labs, Efficacy Assessment, Survival follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. Incidence of adverse events (AEs). 2. Incidence of serious adverse events (SAEs). 3. Incidence of AEs meeting protocol-defined dose-limiting toxicity (DLT) criteria.

7.2 Secondary & Exploratory Endpoints 1. Spleen volume reduction (SVR) at end of Cycle 6 assessed by Blinded Independent Central Review (BICR). 2. Response rate defined as proportion of

participants with SVR $\geq 35\%$ by MRI (preferred) or CT (if MRI is contraindicated and if CT is allowed by local guidelines) assessed by BICR. 3. SVR at end of Cycle 3 and 6 assessed by BICR. 4. Total Symptom Score (TSS) reduction (e.g., $\geq 50\%$ improvement) and Pharmacokinetics (PK) profile of BMS-986158.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring at each clinical visit for dose-limiting toxicities, particularly hematologic AEs (thrombocytopenia, anemia) and gastrointestinal symptoms. **Serious Adverse Events (SAEs):** Reported to the sponsor within 24 hours. **Data Monitoring Committee (DMC):** Internal Safety Review Committee evaluating dose-escalation cohorts.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients receiving ≥ 1 dose of study drug) and Efficacy/Intent-to-Treat populations. **Sample Size Justification:** Targeted enrollment of $N=216$ participants, calculated based on standard Phase 1b dose-escalation algorithms and sufficient powering for preliminary efficacy signals in the Phase 2 expansion cohorts. **Handling of Missing Data:** Standard protocol-defined methods (e.g., Multiple Imputation or Last Observation Carried Forward).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine clinical monitoring by the sponsor. **Audit Trail:** eCRF locking, source data verification, and data clarification mechanisms.

11. Study Identifiers & Governance

Primary ID: CA011-023 **Registry Number:** NCT04817007 **Posting ID:** 606596493447202653 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Study to Assess the Safety and Tolerability of BMS-986158 Alone and in Combination With Either Ruxolitinib or Fedratinib in Participants With Blood Cancer (Myelofibrosis) **Official Regulatory Title:** A Phase 1b/2 Study of BMS-986158 Monotherapy and in Combination With Either Ruxolitinib or Fedratinib in Participants With DIPSS-Intermediate or High Risk Myelofibrosis

12. Clinical Framework (Myelofibrosis Specific)

Primary Condition: Myelofibrosis (Preferred UMLS Name: Myelofibrosis) including Primary Myelofibrosis (PMF), Post-PV MF, Post-ET MF **Diagnosis Threshold:** DIPSS-Intermediate or High Risk **Medical Methodology:** Concomitant BET inhibition and JAK/STAT pathway inhibition **Optimization Target:** Spleen volume reduction and symptomatic improvement

13. Study Procedures & Methodology

Management Pathway: Clinical intervention with targeted oral inhibitors (BMS-986158 $\pm$ Ruxolitinib/Fedratinib) **Education Protocol (HCP):** Sponsor-led investigator site initiation and training **Education Protocol (Patient):** Informed consent process and side-effect management guidelines **Quality Audit Mechanism:** Routine trial monitoring and pharmacovigilance

14. Observation & Follow-Up Schedule

Total Duration: Follow-up for the duration of the study or until primary/study completion (estimated May 2026) **Onsite Visit Cadence:** Pre-defined intervals per cycle throughout treatment **Remote Monitoring Frequency:** As indicated for safety tracking **Checklist Review Interval:** Regular cycle progression assessments

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				